Vector and Parametric Equations of a Line in R^2

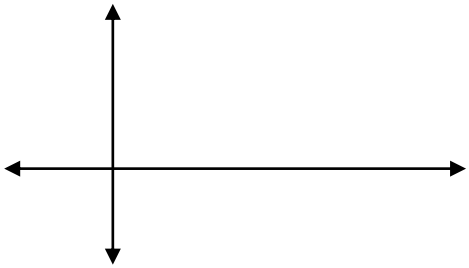
Recall:

To determine an equation of a line in R^2, we would write the equation of a line in the form:

$y = mx + b$

To determine an equation of a line, we need:

1. _____
2. _____



1. Direction

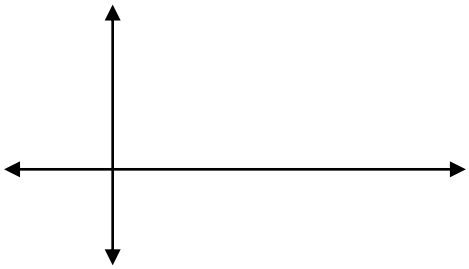
To use vectors, we first define the _____ for the line, $\vec{m}$.

A line in R^2 with the direction vector $\vec{m} = (a,b)$ has a slope of $\frac{b}{a}$.

2. Location

A point on the line has an associated _____, $\overrightarrow{OP}$. Since a line is a set of points, we can consider that line to be a set of position vectors whose endpoints are on the line.

The Vector Equation of a line in R^2



Vector Equation of a line in R^2

$$\vec{r} = \vec{r}_0 + t\vec{m}\,,\ t \in \Re$$

Parametric Equations of a line in R^2

$$(x,y) = (x_0,y_0) + t(a,b),\ t \in \Re$$

Example 1: Given a point $(2,3)$ with a direction vector $\vec{m} = (3,-1)$,

a) Determine the vector equation of this line.

b) Determine the parametric equation of this line.

c) Determine the point on this line that corresponds with the parameter $t=3$.

d) Determine if the point (1,-8) is on the line.

Example 2: Determine a vector equation for the line that is perpendicular to the line $\vec{r} = (1,3) + s(-1,2)$ and passes through the point $(0,5)$.